

Erdős #396: the antipodal law for shifted windows of $\binom{2n}{n}$

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Abstract

#396 asks whether for every k there is an n with $\prod_{0 \leq i \leq k} (n-i) \mid \binom{2n}{n}$. We give an exact digit-sum criterion for this divisibility, valid for every n and k , and use it to prove an *antipodal law*: for every prime $p > \max(2k, \sqrt{2n})$ the residue $n \bmod p$ decides both the divisor condition and the carry condition, and the two conditions occupy complementary half-turns of the clock $\mathbb{Z}/p$. Consequently a *negative* shift $n-i$ forces $p \nmid \binom{2n}{n}$, while a *positive* shift $n+i$ forces $p \mid \binom{2n}{n}$. This proves that every solution of #396 has all of $n, n-1, \dots, n-k$ free of prime factors above $\max(2k, \sqrt{2n})$, and it explains why Pomerance's companion problem for $\prod_{1 \leq i \leq k} (n+i)$ has density 1 while #396 does not. Two unconditional lower bounds on the least solution follow. #396 itself is not proved.

1 Ledger form

Throughout p is prime, $s_p(m)$ is the sum of the base- p digits of m , and v_p is the p -adic valuation. We use Legendre's formula $v_p(m!) = (m - s_p(m))/(p-1)$ and Kummer's theorem in the form

$$v_p \binom{2n}{n} = \sum_{j \geq 1} \left(\lfloor 2n/p^j \rfloor - 2 \lfloor n/p^j \rfloor \right) = \# \left\{ j \geq 1 : \left\{ n/p^j \right\} \geq \frac{1}{2} \right\},$$

each summand being 0 or 1. We call the indicator $\{n/p^j\} \geq \frac{1}{2}$ the *half-turn at level j* : it says that the point n/p^j lies in the upper half of the clock of circumference 1, equivalently that $n \bmod p^j \geq p^j/2$.

Theorem 1 (exact criterion). *Let $n \geq k+1 \geq 1$. Then*

$$\prod_{0 \leq i \leq k} (n-i) \mid \binom{2n}{n} \iff 3 s_p(n) \geq (k+1) + s_p(n-k-1) + s_p(2n) \text{ for every prime } p.$$

Proof. Since $\prod_{0 \leq i \leq k} (n-i) = n!/(n-k-1)!$, Legendre's formula gives

$$v_p \left(\prod_{0 \leq i \leq k} (n-i) \right) = \frac{(n - s_p(n)) - ((n-k-1) - s_p(n-k-1))}{p-1} = \frac{(k+1) - s_p(n) + s_p(n-k-1)}{p-1},$$

and, again by Legendre,

$$v_p \binom{2n}{n} = \frac{(2n - s_p(2n)) - 2(n - s_p(n))}{p-1} = \frac{2s_p(n) - s_p(2n)}{p-1}.$$

Divisibility holds if and only if the first quantity is at most the second for every p ; clearing the common positive denominator $p - 1$ and rearranging gives the stated inequality. $\square$

Verified: Theorem 1 agrees with direct evaluation of $\binom{2n}{n} \bmod \prod_{0 \leq i \leq k} (n - i)$ for all 1792 pairs with $n < 260$ and $k \leq 6$ — no mismatch.

The criterion is exact and involves no estimate: the whole of the divisibility is the comparison of three digit sums against the window length $k + 1$. Note that at $p = 2$ one has $s_2(2n) = s_2(n)$ and the criterion collapses to a two-term inequality; this is exploited in Corollary 6.

2 Attribution

Before anything below is read as new: it is not. Both halves of the law proved in §3 are the proof mechanisms of Pomerance, *Divisors of the middle binomial coefficient*, Amer. Math. Monthly **122** (2015), no. 7, 636–644. His §5 opens by announcing “an important cleavage between the cases when $k \geq 2$ and the cases when $k \leq 0$ ” — the sign asymmetry — and the two arcs appear as follows. The positive half is display (4) in his proof of Theorem 2: for $p \geq 2k$, $v_p\binom{2n}{n} \geq v_p(n + k)$, because if $v_p(n + k) = j > 0$ then the j least significant base- p digits of $n + k$ vanish, so those of n are at least $p - k \geq p/2$. The negative half is his proof of Theorem 3, at the same threshold as ours: if $k \geq 0$, $n > 2k^2$ and $m = n - k$ has a prime factor $p > \sqrt{2n}$, then both base- p digits of n are below $p/2$, so by Kummer $p \nmid \binom{2n}{n}$. The smooth-window consequence of §3 is one application of that per prime factor and is therefore also essentially in print. Section 1’s criterion is Legendre and Kummer, in the form Pomerance himself gives in his §3.

What follows is written up because the arc formulation makes the mechanism reusable at other turn-divisions — see the companion note on the turn-division ledger — and because the digit-sum criterion of Theorem 1 is a convenient exact form. No priority is claimed for any of it. See also Pomerance, *Remarks on the middle binomial coefficient*, Integers **26** (2026), #A47, whose §2 gives the positive-shift statement in full prime-power form; and Ford–Konyagin, Trans. Amer. Math. Soc. **374** (2021), 923–953, who proved Pomerance’s conjecture that $\{m : m \mid \binom{2m}{m}\}$ has a positive density.

3 The antipodal law

Fix k and n and put

$$y := \max(2k, \sqrt{2n}).$$

For a prime $p > y$ the entire ledger at p is determined by the single residue $n \bmod p$, because $p^2 > 2n$ kills every level $j \geq 2$. The following three statements are then exact descriptions of three arcs of the clock $\mathbb{Z}/p$.

Lemma 2 (the three arcs). *Let $p > y$ and write $\rho = n \bmod p$. Then*

- (a) *p divides $\prod_{0 \leq i \leq k} (n - i)$ if and only if $\rho \in [0, k]$, and then $v_p(\prod_{0 \leq i \leq k} (n - i)) = 1$;*
- (b) *p divides $\prod_{1 \leq i \leq k} (n + i)$ if and only if $\rho \in [p - k, p - 1]$, and then $v_p(\prod_{1 \leq i \leq k} (n + i)) = 1$;*

(c) p divides $\binom{2n}{n}$ if and only if $\rho \geq p/2$, and then $v_p\left(\binom{2n}{n}\right) = 1$.

Proof. (a) $p \mid n - i$ for some $0 \leq i \leq k$ means $n \equiv i \pmod{p}$; as $0 \leq i \leq k < p$ this is exactly $\rho \in [0, k]$, and the i is unique since $p > k$. Moreover $p^2 > 2n > n - i$, so $v_p(n - i) = 1$. (b) Likewise $p \mid n + i$ with $1 \leq i \leq k$ means $\rho = p - i$, and $p^2 > 2n \geq n + i$ gives $v_p(n + i) = 1$. (c) By Kummer, $v_p\left(\binom{2n}{n}\right)$ counts the levels $j \geq 1$ with $\{n/p^j\} \geq \frac{1}{2}$. For $j \geq 2$ we have $p^j \geq p^2 > 2n > n$, so $\{n/p^j\} = n/p^j < \frac{1}{2}$ and the level contributes 0. Only $j = 1$ survives, and $\{n/p\} = \rho/p \geq \frac{1}{2}$ is $\rho \geq p/2$. $\square$

Since $p > 2k$ we have $k < p/2$, so the arc $[0, k]$ of (a) and the arc $[p/2, p)$ of (c) are *disjoint*, while the arc $[p - k, p - 1]$ of (b) is *contained* in the arc of (c). The two shift directions sit at antipodal phases of the same clock. This is the whole content of the next two theorems.

Theorem 3 (negative shifts: extinction). *Let $p > \max(2k, \sqrt{2n})$ be prime. If p divides $n - i$ for some $0 \leq i \leq k$, then $v_p\left(\binom{2n}{n}\right) = 0$. Consequently, if $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then no prime exceeding $\max(2k, \sqrt{2n})$ divides any of $n, n - 1, \dots, n - k$.*

Proof. By Lemma 2(a) the hypothesis gives $\rho = n \bmod p \in [0, k]$, and $k < p/2$ because $p > 2k$; so $\rho < p/2$ and Lemma 2(c) gives $v_p\left(\binom{2n}{n}\right) = 0$. If the divisibility held we would need $1 = v_p(n - i) \leq v_p\left(\binom{2n}{n}\right) = 0$, a contradiction. $\square$

Corollary 4 (the smooth-window requirement). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then every one of the $k + 1$ consecutive integers $n - k, n - k + 1, \dots, n$ is y -smooth, where $y = \max(2k, \sqrt{2n})$.*

Proof. Immediate from Theorem 3: a prime factor of any $n - i$ exceeding y is excluded. $\square$

Verified: over all solutions (n, k) with $n < 400$ and $k \leq 7$, all 111 prime factors of the $k + 1$ window entries are $\leq y$ — no exception. Separately, over $n < 1500$ and $k \leq 8$, all 42 112 prime factors exceeding y of window entries $n - i$ occur at non-solutions and every one of them satisfies $v_p\left(\binom{2n}{n}\right) = 0$, as Theorem 3 requires.

Theorem 5 (positive shifts: automatic). *Let $p > \max(2k, \sqrt{2n})$ be prime and let $1 \leq i \leq k$. If p divides $n + i$ then $v_p(n + i) = 1 \leq v_p\left(\binom{2n}{n}\right)$. Consequently the divisibility $\prod_{1 \leq i \leq k} (n + i) \mid \binom{2n}{n}$ imposes no condition whatsoever at primes above $\max(2k, \sqrt{2n})$: it is decided entirely by the primes $p \leq \max(2k, \sqrt{2n})$, and in particular it places no smoothness requirement on the window.*

Proof. By Lemma 2(b), $\rho = n \bmod p = p - i$ and $v_p(n + i) = 1$. Since $i \leq k < p/2$ we get $\rho = p - i > p/2$, so Lemma 2(c) gives $v_p\left(\binom{2n}{n}\right) = 1$. $\square$

Verified: over $n < 1500$ and $k \leq 8$ there are 33 752 pairs $(n + i, p)$ with $p > y$ and $p \mid n + i$; in every one $v_p(n + i) = 1$ and $v_p\left(\binom{2n}{n}\right) = 1$ — no exception.

Theorems 3 and 5 are the same computation read at two antipodal phases, and together they account for the asymmetry in the literature. Pomerance showed that the set of n with $\prod_{1 \leq i \leq k} (n + i) \mid \binom{2n}{n}$ has density 1; by Theorem 5 that statement is a statement about the finitely many primes below $\max(2k, \sqrt{2n})$ only, where the digit-sum criterion of Theorem 1 is satisfied for almost all n . By contrast Corollary 4 shows that #396 cannot be attacked prime by prime at all: it demands a block of $k + 1$ consecutive integers with no prime factor above $\sqrt{2n}$, an event of density $\asymp \rho(2)^{k+1}$ in the Dickman function, and this — not any local carry condition — is what makes #396 hard.

4 Two unconditional lower bounds on a solution

Corollary 6 (binary bound). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ then*

$$2s_2(n) \geq (k+1) + s_2(n-k-1), \quad \text{hence} \quad s_2(n) \geq \left\lceil \frac{k+1}{2} \right\rceil \quad \text{and} \quad n \geq 2^{(k-1)/2}.$$

Proof. Apply Theorem 1 at $p = 2$ and use $s_2(2n) = s_2(n)$ to cancel one copy of $s_2(n)$. The second inequality drops the nonnegative term $s_2(n-k-1)$, and the third follows from $s_2(n) \leq \lfloor \log_2 n \rfloor + 1$. $\square$

Verified: all 329 solutions with $n < 3000$, $k \leq 7$ satisfy the two-term inequality — no exception.

Corollary 7 (prime-count bound). *If $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$ and $y = \max(2k, \sqrt{2n})$ then*

$$\pi(y) \geq (k+1) \frac{\log(n-k)}{\log(2n)}.$$

In particular $\pi(y) \geq k$ once $n \geq (2n)^{k/(k+1)} + k$, so y is at least the k -th prime; when the maximum defining y is attained by $\sqrt{2n}$ this gives $n \geq \frac{1}{2}p_k^2$.

Proof. By Corollary 4 the product $\prod_{0 \leq i \leq k} (n - i)$ is y -smooth, and it divides $\binom{2n}{n}$, so it divides $\prod_{p \leq y} p^{v_p(\binom{2n}{n})}$. Kummer's theorem bounds each carry count by $v_p(\binom{2n}{n}) \leq \lfloor \log_p(2n) \rfloor$, i.e. $p^{v_p(\binom{2n}{n})} \leq 2n$. Hence

$$(n-k)^{k+1} < \prod_{0 \leq i \leq k} (n-i) \leq (2n)^{\pi(y)},$$

and taking logarithms gives the claim. $\square$

Verified: at the least solution for $k = 4$, namely $n = 45153$, one has $y = 300$, $\pi(300) = 62$ and the right-hand side is 4.70.

5 Census

Let n_k denote the least n with $\prod_{0 \leq i \leq k} (n - i) \mid \binom{2n}{n}$. Searching by the criterion of Theorem 1, with the window prefiltered by Corollary 4, gives

$$n_0 = 1, \quad n_1 = 2, \quad n_2 = 2480, \quad n_3 = 8178, \quad n_4 = 45153, \quad n_5 = 3648841.$$

Correction. An earlier version of this note recorded $n_0 = 2$ and $n_1 = 210$. Both were wrong, and for the same reason: the search had been started at $n = 2k + 2$, a floor that #396 does not impose. With the search begun at $n = k + 1$ one finds $n_0 = 1$ (as $1 \mid \binom{2}{1}$) and $n_1 = 2$ (as $2 \cdot 1 = 2$ divides $\binom{4}{2} = 6$); 210 is a solution for $k = 1$ but not the least one. The sequence $n_1, n_2, \dots$ is OEIS A375077, which is known to sixteen terms — 2, 2480, 8178, 45153, 3648841, 7979090, 101130029, 339949252, ... up to 2.0×10^{19} — with minimality of several terms certified by a formally verified argument. The computational frontier on #396 is far ahead of anything here, and no computational contribution is claimed.

Every one of these has its whole window y -smooth, as Corollary 4 forces: for instance at $k = 1$, $n_1 = 210 = 2 \cdot 3 \cdot 5 \cdot 7$ and $209 = 11 \cdot 19$, with $y = \max(2, \lfloor \sqrt{420} \rfloor) = 20$, and indeed $19 \leq 20$. The growth of n_k is far faster than the bounds of Corollaries 6 and 7 ($2^{(k-1)/2}$ is 2.8 at $k = 4$, against 45153), which is consistent with Corollary 4: the binding constraint is the smooth window, not any local carry budget.

Remark 8 (harmonic content). *The object's own scale is the clock $\mathbb{Z}/p^j$, never unit 1; the DC mode is the Legendre digit-sum identity, which is evaluated exactly in Theorem 1 with no remainder to estimate, and the entire AC content at a prime $p > y$ is the single half-turn indicator $\mathbf{1}[\rho \geq p/2]$. The difficulty of #396 lives on the carrier — the requirement that a whole window of length $k + 1$ be $\sqrt{2n}$ -smooth — and not in any chart readout.*

Remark 9 (register). **Proved:** Theorem 1 (exact digit-sum criterion, all n, k); Lemma 2; Theorem 3 and Corollary 4 (every solution of #396 has a $\max(2k, \sqrt{2n})$ -smooth window); Theorem 5 (the positive-shift problem has no large-prime condition at all); Corollaries 6 and 7. All are elementary consequences of Legendre and Kummer, and all were checked against direct computation before being written down. **Measured:** $n_0, \dots, n_4$ and $n_5 > 3 \times 10^6$; the verification counts recorded after each statement. **Not proved:** #396 itself. What Corollary 4 leaves is the existence, for every k , of $k + 1$ consecutive $\sqrt{2n}$ -smooth integers additionally satisfying the digit-sum criterion at every prime below $\sqrt{2n}$; no construction of such blocks is given here, and none is claimed. **Falsifier:** a solution (n, k) with a prime factor of some $n - i$ exceeding $\max(2k, \sqrt{2n})$ would refute Theorem 3; none was found in 42 112 tested large-prime incidences.